

Gazebo Simulator For Part2

Purpose

Beware the simulator is not the exact same as the final test. Don't treat simulators as the final exam.

The purpose of the simulator is to help students who want to run simulation temporarily, and do not have access to the lab devices.

Setting up

Download files

Go to Canvas-Files-Customed_Simulator, download all the files (including .sdf file) in to your own directory.

e.g. /home/user/report2_sim

Unset real turtlebot4

```
unset -turtlebot
```

Set environment variables

Set gazebo resource directory

```
export GZ_SIM_RESOURCE_PATH=$GZ_SIM_RESOURCE_PATH:{your_path_to_sdf}
```

Disable GPU mode

Optional: if any error with GPUs:

```
export LIBGL_ALWAYS_SOFTWARE=1
```

Run simulator

Caution: No file extension for sdf file parameter

Open a terminal under your simulator file. Input:

```
ros2 launch turtlebot4_ignition_bringup turtlebot4_ignition.launch.py world:=report2_simulator
```

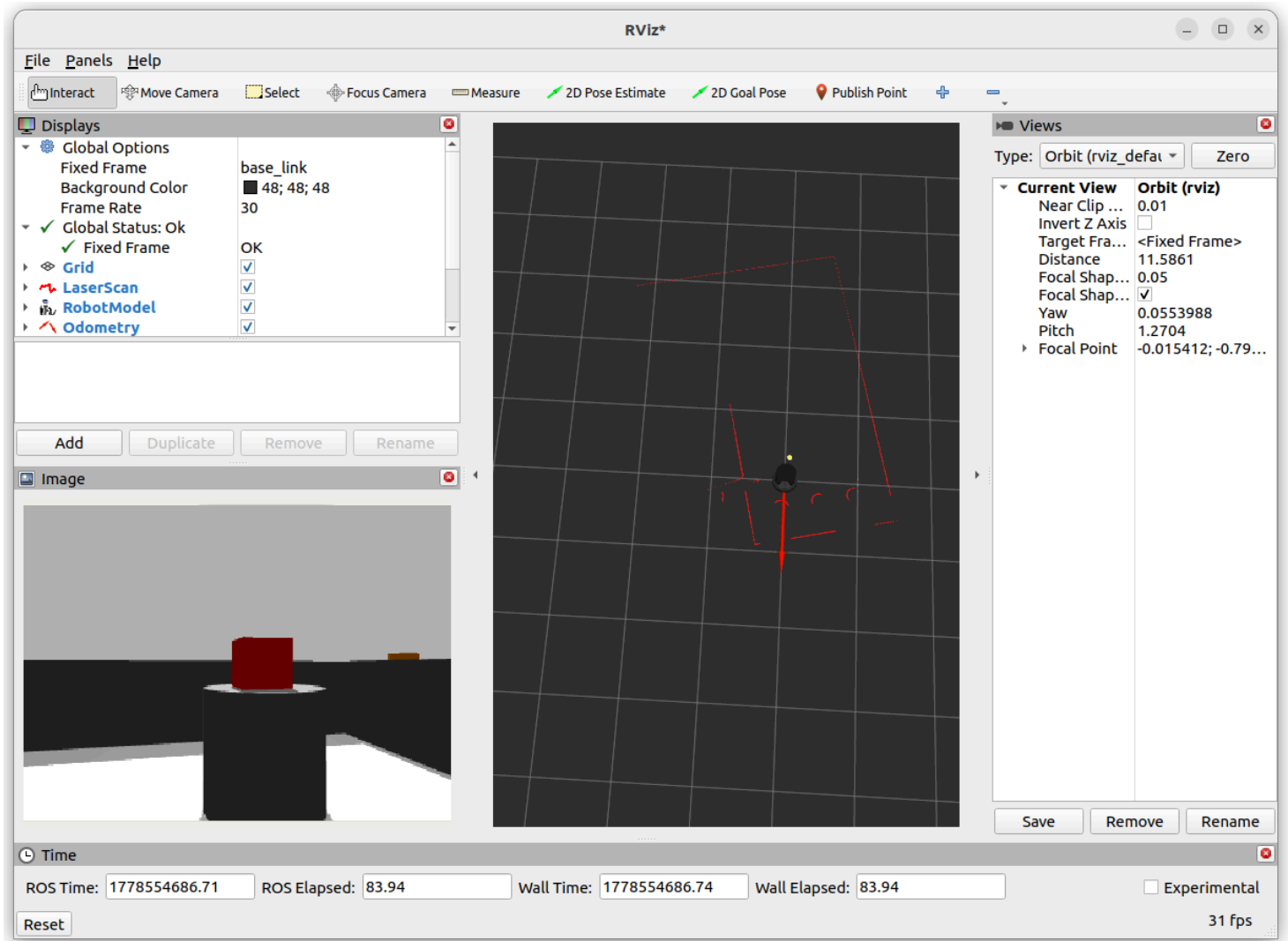
When the environment shows, you need to press start button instantly after the environment shows. Otherwise, the remote control components fail when they can not connect to the simulated robot.

Teleoperation

```
ros2 run teleop_twist_keyboard teleop_twist_keyboard
```

Visualize with Rviz

```
ros2 run rviz2 rviz2
```



Change the position of the target and obstacles

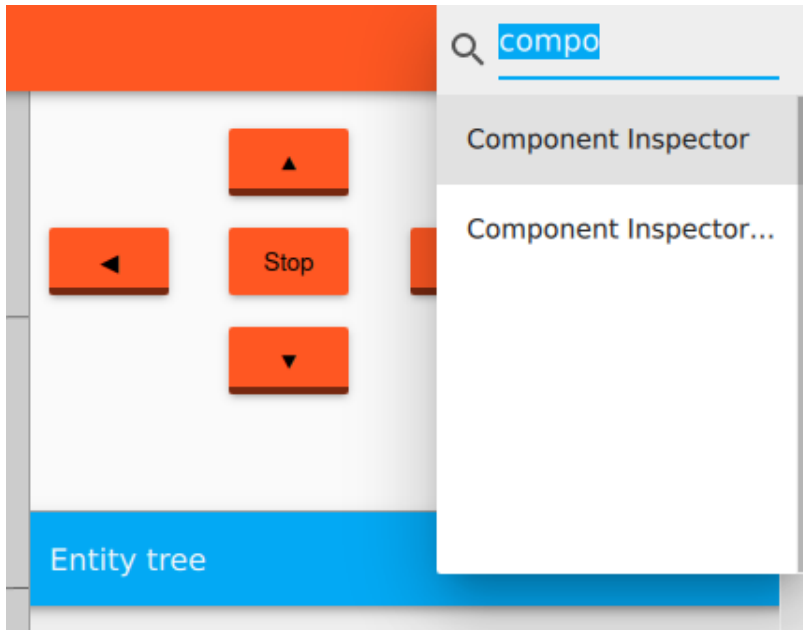
You need to change the position of the obstacles and the target manually as follows:

1. press start button to pause simulator

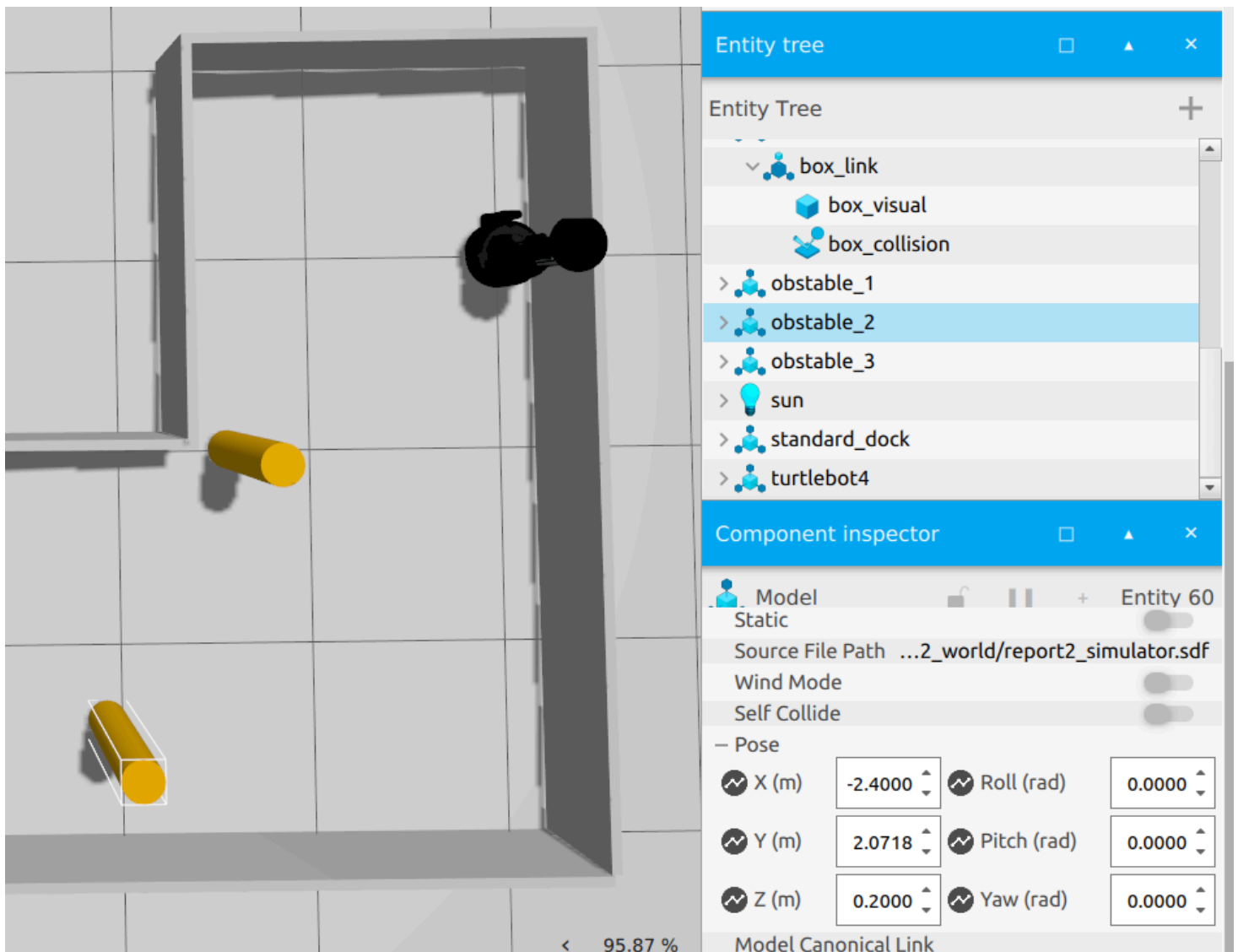


2. press three dots on the right top window

3. search 'Entity tree' and 'Component inspector', choose the search result to add the panel into right sidebar



4. choose target or obstacle 1, 2, 3 at entity tree, and then change the pose in the simulated environment or Component inspector



- the position of target should be in one of the $(0.6, 4.2, 0.24)$, $(0.6, 4.7, 0.24)$, $(0.6, 5.2, 0.24)$, z-axis should be fixed as 0.24.
- use the up-down button to add 1m to xy axis quickly for obstacles. Or you could input customized random location.